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Palaeomagnetic Results for the Permian of South America and a Comparison with the African and Australian Data

B. J. J. Embleton

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Summary

Palaeomagnetic data are summarized for two collections of red-beds from the Middle Paganzo formation (Permian age) of La Rioja and San Juan Provinces, north-west Argentina. The thicker succession sampled in La Rioja allows us to trace the movement of the south palaeomagnetic pole which shifted during the Permian from 59·5° S, 2·5° W to 74° S, 52° W. The red-beds from San Juan yield a south palaeomagnetic pole situated at 63° S, 4° W. A basalt which occurs near the base of the La Colina formation (in the Paganzo type area, and directly equivalent to the middle Paganzo) yields a south virtual palaeomagnetic pole at 50° S, 3° E. The results are entirely consistent with previously published palaeomagnetic data for the Permian and Permo-Carboniferous of South America.

1. Introduction

Palaeomagnetic investigation of South American rock formations commenced in 1956. In 1958, K. M. Creer extended the work to north-west Argentina. The initial project was intended as a preliminary palaeomagnetic survey which could also be used for planning future investigations. Several review papers have been published **(1)**, **(2)**, **(3)** describing the results as they became available and recently, a comprehensive paper dealing with all published and the previously unpublished data have been presented. **(4)**

During the last three years, a detailed palaeomagnetic study of Permian rocks from South America has been made to assist in explaining the Permian geomagnetic field relative to that continent. Until recently, the polar wandering curve for South America has been drawn based on data provided mainly by Creer **(2)**, **(3)**. Only one palaeomagnetic pole of Permian age, however, had been determined. Corroboration of this pole position and its validity for the whole of the Permian period was desirable since the African and Australian palaeomagnetic data have been interpreted **(5)**, **(6)** such that the palaeomagnetic record of South America appears to be incomplete. In fact, the similarity between the African and Australian data for the Upper Palaeozoic is striking **(7)**, the resulting continental reconstruction supporting that of King's **(8)** earlier work. In the light of the new results for South America presented here, however, a reconstruction is suggested based on Permo-Carboniferous data for each continent. Palaeomagnetic study of several Triassic formations of South America **(9)** show that it was adjacent to Africa at least until the end of the Triassic although by this time, Australia, Antarctica and possibly India too, had drifted away from these

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two continents (**10**). Thus, the geocentric dipole hypothesis applied in palaeomagnetism requires that the Permian south poles of Africa and South America were coincident when the two continents are reconstructed to the positions they occupied relative to one another in the Permian period.

2. Permian data for South America

Palaeomagnetic collections were made in 1967 and 1969 in the Paganzo Basin situated in north-west Argentina. The Middle Paganzo formation (Paganzo II) is considered Permian in age (**11**), (**12**), and consists of massively bedded, brick red, fine to medium grained sandstones. In the type area near Paganzo village (Province of La Rioja) the formation attains a thickness of 700 m and in the lower part of the succession, volcanic intercalations are present. These usually take the form of tuffs and the only horizon suitable for palaeomagnetic study is a basalt which occurs near the base of the Middle Paganzo formation, known in the type areas as the La Colina formation. The palaeogeographic and geological evolution of the Basin has been recently described by two Argentine geologists, Azcuy & Morelli (**11**). Rock collections were made at three localities, (a) the red-beds of Los Colorados, La Rioja Province (29·5° S, 67° W), (b) the red-beds of Huaco, San Juan Province (30° S, 68·5° W), and (c) the basalt in the type area, Paganzo, La Rioja Province (30° S, 67° W). The localities are marked in Fig. 1.

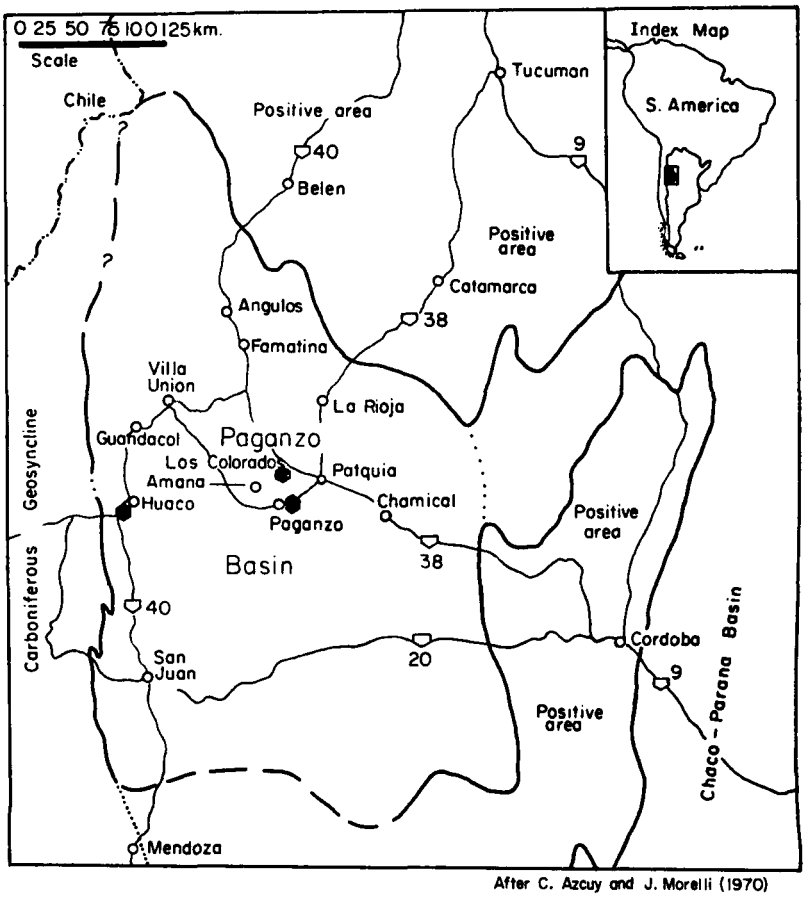


FIG. 1. Palaeogeographic units existing during the deposition of the Paganzo Group. The hexagonal points indicate the sampling localities.

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2.1 *Summary of palaeomagnetic results from Los Colorados*

Fifty-three hand samples of brick-red, compact sediments were collected from beds dipping uniformly westwards at 12°. The directions and intensities of remanent magnetization were measured in Newcastle upon Tyne, England. All specimens were subjected to thermal demagnetization in zero field conditions to test the stability of the remanence and erase secondary components that had been subsequently acquired. The mean sample directions of magnetization before and after thermal treatment are summarized in Table 1 and shown in Fig. 2. Fig. 3 demonstrates the streaking that existed prior to thermal treatment, and indicates that the secondary component of magnetic remanence was acquired under the influence of the present

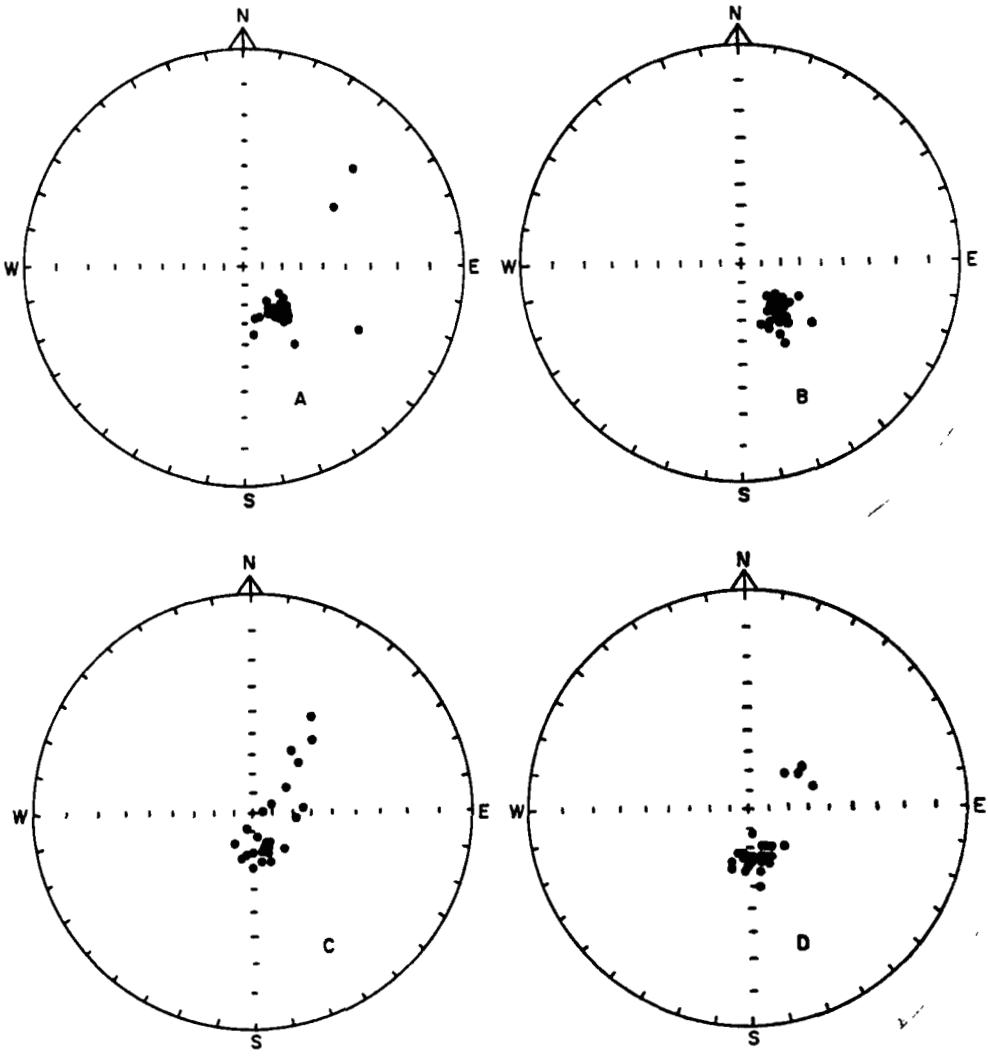


FIG. 2. Middle Paganzo, Los Colorados; sample-mean directions of remanent magnetization referred to the bedding planes. A and B shows sample mean directions of magnetization before and after thermal treatment for samples collected from the lower beds, C and D show the directions of magnetization of the upper beds before and after treatment. Closed circles indicate (positive) downward inclinations, that is, reversed polarizations for the Southern Hemisphere.

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Table 1

Summary of palaeomagnetic results from the Middle Paganzo, Los Colorados. Unit weight for calculation of mean directions of remanent magnetism was assigned to each sample

	Mean direction of remanent magnetization					South Pole			Stratigraphic position
	Dec.	Incl.	No. of samples	No. of specimens	α_{95}	Lat.	Long.	α_{95}	
n.r.m.	135·5°	+59·0°	26	104	7·0°	54·5° S	6° W	8·5°	lower beds
470 °C	144·0°	+57·0°	26	78	2·5°	59·5° S	2·5° W	3·5°	
n.r.m.	157·0°	+73·5°	27	108	5·0°	54·5° S	45° W	9·5°	upper beds
470 °C	174·5°	+63·0°	23	69	3·5°	74° S	52° W	4·5°	

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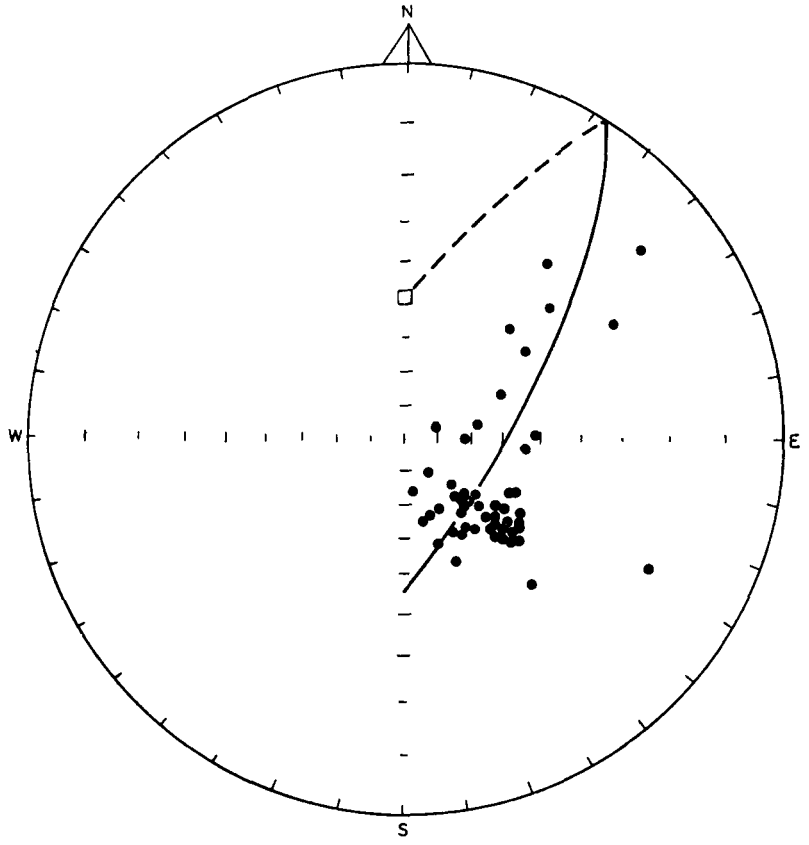


FIG. 3. Sample-mean directions of natural remanent magnetization of samples from the Middle Paganzo, Los Colorados, referred to the present horizontal, demonstrate the streaking that existed prior to thermal cleaning. A great circle through these points is shown to pass through the present dipolar field position. Partial remagnetization in the direction of the present field has occurred in some cases subsequent to their acquisition of the primary remanence.

geomagnetic field. It was eliminated in all but four samples which are assumed to have acquired a more stable secondary component. Samples were collected at regular intervals over a thickness of approximately 300 m. About midway in the succession the nature of the magnetic remanence alters. Samples from the lower part have relatively high intensities of natural remanence, the intensity spectrum lying between 10 μG and 100 μG with a geometrical mean of 38·1 μG. The directions of remanent magnetization of these 26 samples after thermal treatment are approximately parallel and all exhibit reversed polarity (see Fig. 2). The range of intensities of the 27 samples from the upper part of the succession is 1 μG to 10 μG with the geometrical mean at 3·48 μG. After thermal treatment their directions of remanent magnetization are likewise approximately parallel (and reversed) but appear to form a group distinct from the population obtained from the lower part of the succession.

South palaeomagnetic poles calculated from the two distributions are given in Table 1. Table 2 shows how the magnetic vector changes direction (and also the South Pole) as the succession is traversed. There is seen to be a significant change in the direction of the magnetic vector between samples 1643 and 1644 although the succession is conformable throughout. The difference between the two mean directions is interpreted as being due to a polar shift during the late Permian.

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Table 2

Change in direction of the magnetization vector as the Los Colorados succession is traversed

	Mean direction of remanent magnetization				South Pole			Stratigraphic position
Sample No.	Dec.	Incl.	No. of samples	α_{95}	Lat.	Long.	α_{95}	
1618–1623	134·0°	+57·0°	6	7·5°	51·5° S	2° W	8·5°	lower beds
1624–1628	148·5°	+53·0°	5	7·0°	63° S	6° E	8·0°	
1629–1633	151·0°	+55·0°	5	7·0°	65° S	1° E	8·5°	
1634–1638	143·0°	+61·5°	5	5·0°	58° S	11° W	7·0°	
1639–1643	145·0°	+58·5°	5	4·5°	60° S	5° W	5·0°	
1644–1648	171·5°	+65·0°	5	8·0°	71° S	47° W	12·0°	upper beds
1649–1653	184·0°	+64·5°	5	7·0°	72·5° S	64·5° W	9°5·	
1665–1670	165·0°	+62·0°	5	11·0°	72° S	29° W	17·0°	
1671–1674	172·5°	+59·0°	3	16·5°	78° S	42° W	22·0°	
1677–1681	175·5°	+62°5·	5	7·5°	75·5° S	54° W	11·0°	

Table 3

Summary of palaeomagnetic results from the Middle Paganzo, Huaco. For the calculation of the mean directions of remanent magnetism, unit weight was assigned to each sample

	Mean direction of remanent magnetism				South Pole		
Dec.	Incl.	No. of sample	No. of specimens	α_{95}	Lat.	Long.	α_{95}
158·5°	+58·0°	9	26	7·5°	70° S	12° W	10·0°
154·5°	+57·0°	9	19	5·5°	67·5° S	6° W	7·0°
148·0°	+56·5°	9	19	7·5°	63° S	4° W	9·5°

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2.2 *Summary of palaeomagnetic results from Huaco*

Nine samples of red sediments were collected from steeply dipping beds (50° to N 285° E) in Huaco Gorge, Province of San Juan. The results summarized in Table 3 and shown in Fig. 4 are given before treatment and after thermal treatment at 450 °C and 650 °C. The remanent magnetization was found to be stable and contained only weak secondary components. We may note that as the temperature, at which thermal demagnetization was carried out, was increased, the computed south pole moved systematically from 70° S, 12° W to 63° S, 4° W, that is, to a position close to the palaeomagnetic pole calculated from the lower beds of the Los Colorados succession. All specimens exhibited reversed polarization.

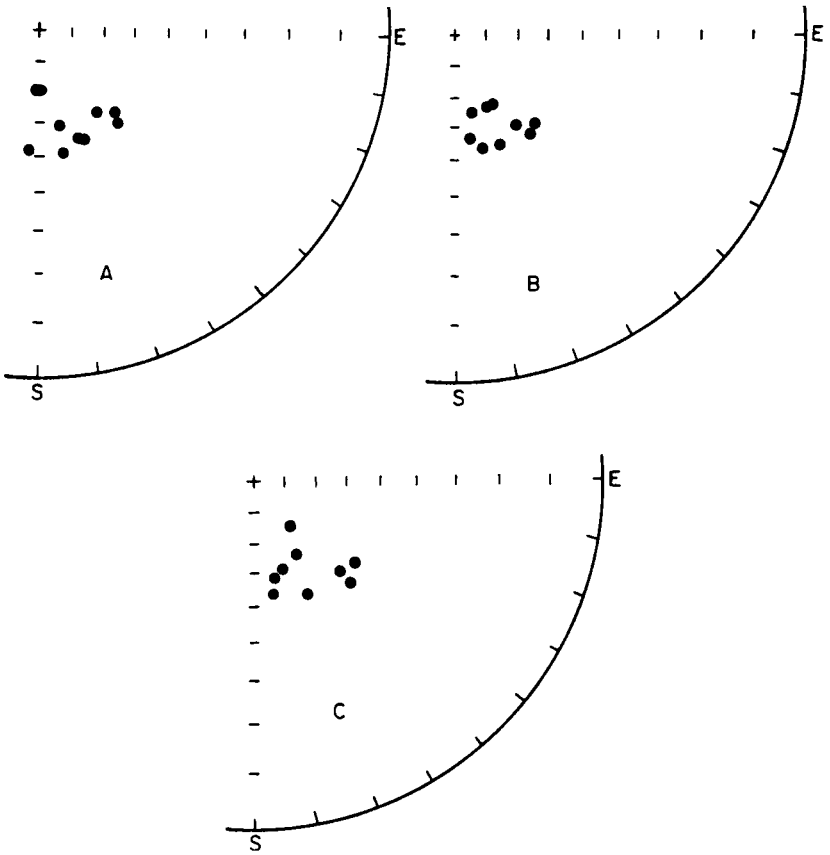


FIG. 4. Middle Paganzo, Huaco: sample-mean directions of remanent magnetization referred to the bedding planes. A shows the directions before treatment, B after thermal demagnetization at 450 °C and C after demagnetization at 650 °C.

2.3 *Summary of palaeomagnetic results from the La Colina Basalt*

Eight samples were collected at two sites (separated by about 500 m) from a basalt which occurs stratigraphically near the base of the La Colina formation (equivalent to the Middle Paganzo) in the type area near Paganzo village (30° S, 67° W). The basalt is approximately 10 m thick and dips steeply to the south-east. The dip was measured in the red sediments above and below the lava flow (strike, N 37° E; dip, 62°).

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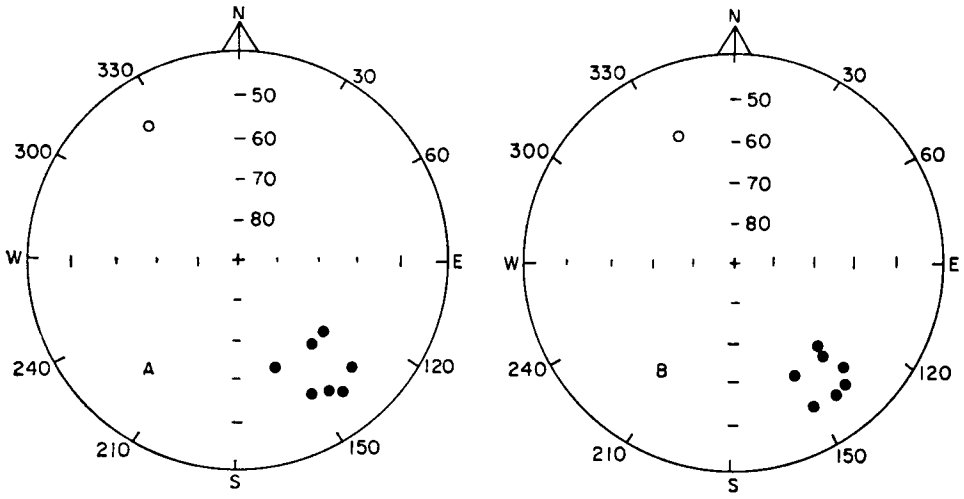


FIG. 5. La Colina basalt, Paganzo type area: sample-mean directions of remanent magnetization referred to the bedding plane (a) before cleaning and (b) after alternating field cleaning in a peak field of 200 oersteds. One sample (open circle, upward direction) is normally magnetized.

The natural remanent magnetism of four discs from each sample was measured in the palaeomagnetic laboratory of the Faculty of Natural Sciences, the University of Buenos Aires, Argentina. All samples showed internal consistency of directions of magnetic remanence; the distribution of between-sample directions may be seen in Fig. 5 (a). One sample was magnetized normally although the remainder were reversed.

One specimen was chosen from each sample and treated in alternating magnetic fields up to 510 oersteds (peak value). Each of the eight samples were shown to be stable. The directions of the contained magnetism remained almost unchanged although a smooth decrease in intensities was observed in all reversely magnetized specimens. The palaeomagnetic results are summarized in Table 4 and the sample-mean directions of the magnetic remanence after alternating field treatment are shown in Fig. 5 (b). A peak field of 200 oersteds was chosen as adequate to eliminate the apparently weak secondary components of remanence. Two further specimens from each sample were chosen for treatment in this field; to serve as a check on the measurements made on the pilot specimens and to provide data from which sample-means were obtained.

3. Discussion of results

Palaeomagnetic results from the Middle Paganzo red-beds have been demonstrated to be repeatable. The thicker succession sampled in Los Colorados permits us to trace the movement of the south palaeomagnetic pole during the Permian period, *viz.* to the south and west of the Permo-Carboniferous polar region. In Fig. 6 is shown the three south poles (numbers 2, 3 and 5) determined from the Middle Paganzo formation sampled at Huaco and Los Colorados. The age of the formation is Permian, the lower beds of Los Colorados and the beds of Huaco yielding poles (numbers 2 and 3) very close to the coincident poles (number 4) of the Piaui formation of Brazil and the Permian red-beds of Argentina. The pole computed from the upper beds of Los Colorados (number 5) is seen from Fig. 6 to lie close to the three Triassic poles determined from basalt successions **(9)**, **(13)** of Nihuil (number 7, Lower Triassic), Cuesta los Teneros (number 8, Triassic) and Uspallata

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Table 4

Summary of the palaeomagnetic results of the La Colina basalt, Paganzo type area

Sample	Direction of remanent magnetism				Virtual South Pole		
	Dec.	Incl.	No. of Specimens	α_{95}	Lat.	Long.	Polarity
2042	335°	−57°	3	7·5°	68° S	4° W	N
2043	134°	+53°	3	1·5°	51° S	5° E	R
2044	143°	+49°	3	2·5°	58° S	13° E	R
2045	136°	+58°	3	2·5°	53° S	2° W	R
2058	138°	+50°	3	3·5°	54° S	11° E	R
2059	152°	+59°	3	5·0°	65° S	8° W	R
2060	151°	+50°	3	2·5°	65° S	12° E	R
2061	134°	+68°	3	1·5°	52° S	6° W	R

Mean Direction of remanent magnetization				South Pole Position		
Dec.	Incl.	No of Samples	α_{95}	Lat.	Long.	α_{95}
143°	+55°	8	4·5°	59° S	3° E	5·5°

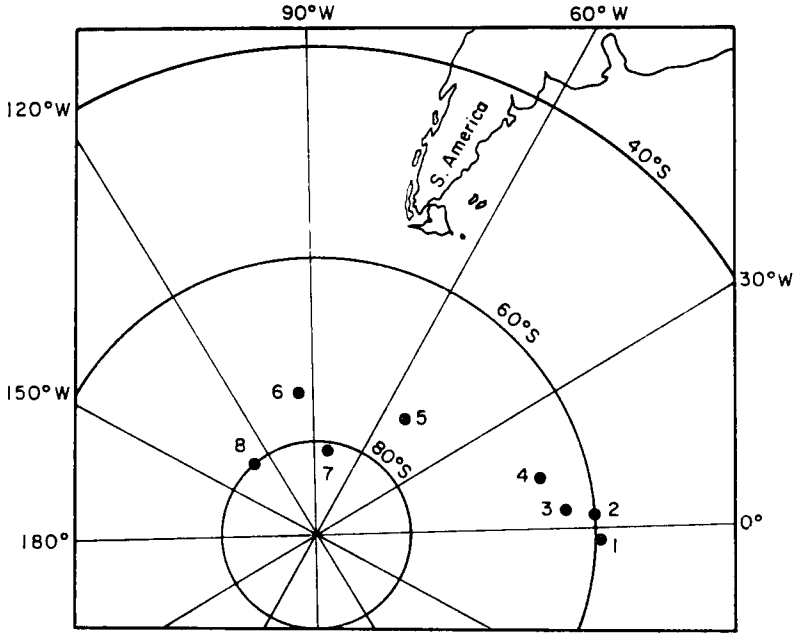


FIG. 6. The four new palaeomagnetic south poles presented in this article: 1, La Colina basalt; 2, Middle Paganzo, Los Colorados (lower beds); 3, Middle Paganzo, Huaco; 5, Middle Paganzo, Los Colorados (upper beds) are shown in relation to other Permo-Carboniferous and Triassic south palaeomagnetic poles of South America; 4, Piaui formation, 65° S, 13° W, $\alpha_{95} = 10°$ and Permian red-beds from Argentina, 65° S, 13° W, $\alpha_{95} = 6°$; 6, Upper Triassic lavas (Upsallata), 75° S, 94° W, $\alpha_{95} = 14°$; 7, Lower Triassic lavas (South Nihuil) 81° S, 78° W, $\alpha_{95} = 6°$; 8, Triassic lavas (Cuesta los Teneros), 80° S, 132° W, $\alpha_{95} = 10°$.

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(number 6, Upper Triassic) from the Province of Mendoza, western Argentina. The basalts are predominantly normally magnetized.

Since the succession that was sampled in Los Colorados is conformable throughout, it would appear that we now have a reasonably complete picture of the Permian geomagnetic field relative to South America.

The palaeomagnetic results from the La Colina basalt lend further support to the observations so far. The stability of the magnetic remanence to alternating field cleaning indicates that the remanence is primary. The mean direction of magnetization obtained after treatment is taken to represent the direction of the ambient geomagnetic field when the lava flow was formed. It is thus no more than a point record of the direction of the palaeo-field and is subject to the effect of secular variation. The computed virtual south pole, however, supplies useful corroboration of the palaeomagnetic results obtained from red-beds formations (pole number 1, Fig. 6).

4. Comparison of Permian Data for South America, Africa and Australia

The above results indicate that the polar region, relative to South America, which was initially defined by Creer (2) for the period Permo-Carboniferous, existed for

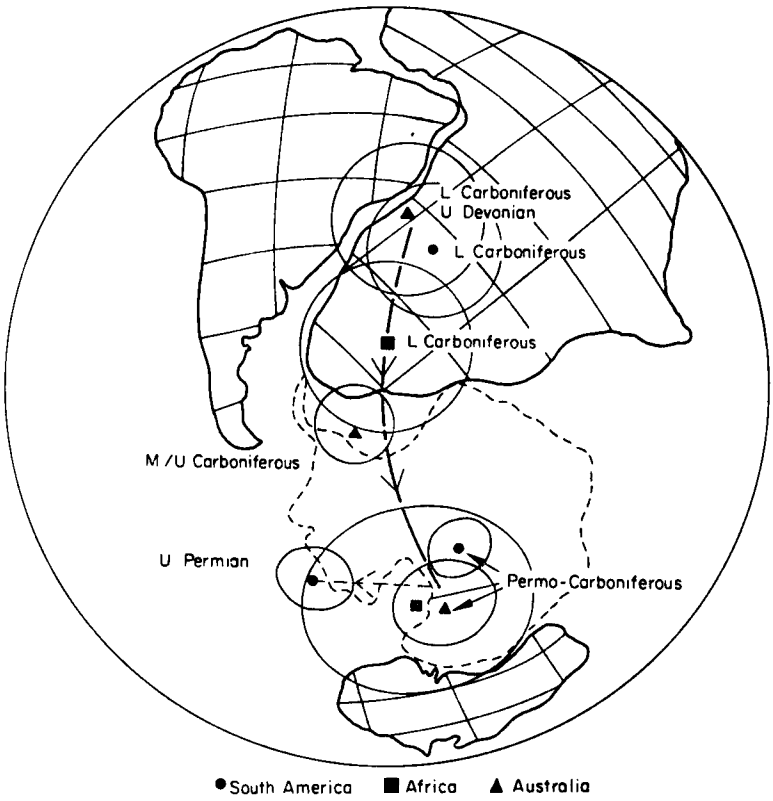


FIG. 7. Palaeomagnetic reconstruction of South America, Africa and Australia obtained by placing the Upper Palaeozoic south poles on a common polar wander curve and superimposing the Permo-Carboniferous poles for each continent. (For Africa, the mean of the Upper Carboniferous and Lower Permian poles is used). Consideration has also been taken of morphological constraints, hence the Permo-Carboniferous poles are not exactly coincident—but they are not significantly different from one another, (circles of confidence are drawn at the 95 per cent probability level of the mean). The palaeomagnetic data on which the reconstruction is based is listed in Table 5.

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Table 5

List of South Palaeomagnetic Poles on which the reconstruction in Fig. 7 is based

	Lower Carboniferous Upper Devonian				Middle/Upper Carboniferous				Permo Carboniferous				Upper Permian			
	Lat.	Long.	α_{95}	N	Lat.	Long.	α_{95}	N	Lat.	Long.	α_{95}	N	Lat.	Long.	α_{95}	N
S. America (SA)	28° S	326° E	(9°)	—	—	—	—	—	63° S	353° E	4°	4	74° S	308° E	(5°)	—
Africa (Af)	26·5° S	26·5° E	(11°)	—	—	—	—	—	40° S	64·5° E	16·5°	2	—	—	—	—
Australia (Au)	76° S	2° E	11°	4	73° S	147° E	(4·5°)	—	46° S	133° E	7°	6	—	—	—	—

Source of Data: S. America; references **(2)**, **(14)** and this paper.
Africa; reference **(7)**.
Australia; reference **(5)**.

Note: N is given only when several poles are averaged to obtain a mean. α_{95} in all other cases is the circle of confidence of the mean at the 95 per cent probability level quoted from the original palaeomagnetic study. The Permian result from the La Colina basalt (this paper and reference **(13)** is not included in the mean Permo-Carboniferous pole quoted for South America.

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much of the Permian. Polar movement probably took place during the late Permian, the South Pole shifting towards the Triassic position where it remained fixed until the end of the Triassic period (9), (13). When this result is compared with the African data (14), (15), the African Lower Permian pole lies significantly to the east of the South American Permo-Carboniferous pole. Assuming that palaeomagnetic poles are true records of the Permian geomagnetic field relative to South America and Africa, that is, that (i) sampling has adequately removed secular variation effects, (ii) magnetic cleaning has removed secondary components of magnetic remanence, and (iii) the Permian field was that of a geocentric dipole, then the difference between the two results cannot yet be resolved. However, if the Upper Carboniferous and Lower Permian poles for Africa are combined, it is suggested that a better estimate of the palaeomagnetic pole for this period is obtained, and an Upper Palaeozoic reconstruction is made possible based on Permo-Carboniferous data from both South America and Africa. Whether the poles ought to be combined or not has been discussed recently (7).

The position of the Permian palaeomagnetic pole relative to Australia is well documented (16) and reconstruction of Australia to its Gondwana location is mainly dependent upon whether this pole is superimposed with the Permo-Carboniferous pole for Africa or the Lower Permian pole for Africa. The mean south pole quoted for Australia (Table 5) is in fact based on an Upper Carboniferous and Permian population, (5) previously referred to as the Kiaman (7) pole for Australia. The reconstruction shown in Fig. 7 is based on the Permo-Carboniferious results for South America, the combined Permian and Carboniferious results for Africa and the Kiaman results for Australia. The position of Antarctica between Australia and Africa is shown dotted, its orientation determined from sea-floor spreading (17) and morphological (18), (19) studies. The palaeomagnetic data used in the reconstruction is listed in Table 5. The fit of the three continents varies little in detail from previous proposals (for example, see references (6), (7), (10), (19) and (20).

5. Conclusions

Palaeomagnetic study of several Permian rock collections from South America has failed to yield a south palaeomagnetic pole which would support the African Lower Permian data when the two continents are considered in their pre-drift situations. The South American data suggests in fact, that the South Pole made a south-westward discursion during the late Permian towards the coincident Triassic polar regions of South America and Africa.

All Permian formations studied so far from South America are reversely magnetized; the inclinations of the magnetic vectors are reversed with respect to the present geomagnetic field inclinations at the sampling sites. The Piaui formation is also reversely magnetized. The magnetizations are believed to date from the Kiaman Magnetic Interval which was first detected and defined from palaeomagnetic studies on Australian Upper Carboniferous and Permian formations (21), (22). A polarity change from ' normal ' to ' reversed ' was demonstrated by Irving (22), in the Upper Carboniferous section of Hunter Valley, New South Wales. Reversed geomagnetic polarity then persisted for almost the whole of the Permian period, the next major polarity change (to ' normal ') occurring in the Lower Triassic. The presence of the Kiaman Magnetic Interval in Permian formations of the U.S.A., the U.S.S.R. and western Europe has been indicated, and the possibility of inter-continental correlation has been discussed by McElhinny (23).

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References

(1) Creer, K. M., 1964. A reconstruction of the continents for the Upper Palaeozoic from palaeomagnetic data, *Nature Lond.*, **203,** 1115–1120.

(2) Creer, K. M., 1964. Palaeomagnetism and the results of its application to South American rocks, *Bol. Paranaense Geografia*, 93–138.

(3) Creer, K. M., 1965. Palaeomagnetic data from the Gondwanic continents, *Phil. Trans. R. Soc. Lond., Ser. A*, **258,** 27–40.

(4) Creer, K. M., 1970. A palaeomagnetic survey of South American rock formations, *Phil. Trans. R. Soc. Lond., Ser. A* (in press).

(5) Briden, J. C., 1967. Recurrent continental drift of Gondwanaland, *Nature*, **215,** 1334–1339.

(6) Creer, K. M., Embleton, B. J. J. & Valencio, D. A., 1969. Comparison between the Upper Palaeozoic and Mesozoic palaeomagnetic poles for S. America, Africa and Australia, *Earth Planet. Sci. Lett.*, **7,** 288–292.

(7) McElhinny, M. W., Briden, J. C., Jones, D. L. & Brock, A., 1968. Geological and geophysical implications of palaeomagnetic results from Africa, *Rev. Geophys.*, **6,** 201–237.

(8) King, L. C., 1962. *The morphology of the Earth*, Oliver and Boyd, Edinburgh.

(9) Creer, K. M., Embleton, B. J. J. & Valencio, D. A., 1969. Triassic and Permo-Triassic data for S. America, *Earth planet. Sci. Letts.* (in press)

(10) Valencio, D. A., Embleton, B. J. J. & Vilas, J. F., 1970. Reconstruccion y evolucion del continente Gondwanico sobre la base de datos palaeomagneticos y de propagacion de los fondos de los oceanos, *Rev. Assoc. Geol. Arg.* (in press).

(11) Azcuy, C. & Morelli, J., 1970. Geologia de la Comarea Paganzo-Amana, Provincia de la Rioja, *Rev. Assoc. Geol. Arg.*, (in press).

(12) Bodenbender, G., 1912. Parte Meridional de la Provincia de La Rioja y regiones limitrofes, *Anal. Min. Agric. Secc. Geol. Mineral Min.*, vol. VII, No. 3.

(13) Valencio, D. A., 1969. El paleomagnetismo de alguinas magmatitas del Triasico superior, Grupo Cacheuta, *Rev. Assoc. Geol. Arg.*, **XXIV,** 191.

(14) Embleton, B. J. J., 1970. Nuevos estudios paleomagneticos del neopalaeogoico (Paganzo II) de la Republica Argentina, *Rev. Assoc. Geol. Arg.*, **XXV,** 101–110.

(15) Embleton, B. J. J., 1970. Resultados paleomagneticos del basalto de la base de la formacion La Colina (Paganzo Medio), La Rioja, Republica Argentina, *Rev. Assoc. Geol. Arg.* (in press).

(16) Irving, E., 1964. *Palaeomagnetism—and its application to geological and geophysical problems*, John Wiley and Sons, New York.

(17) Vilas, J. F. & Valencio, D. A., 1970. Palaeogeographic reconstruction of the Gondwanic continents based on palaeomagnetic and sea-floor spreading data, *Earth Planet. Sci. Letters.*, **7,** 397–405.

(18) Sproll, W. P. & Dietz, R. S., 1969. Morphological continental drift fit of Australia and Antarctica, *Nature*, **222,** 345–348.

(19) Smith, A. G. & Hallam, A., 1970. The fit of the southern continents, *Nature*, **225,** 139–144.

(20) Creer, K. M., 1967. *A synthesis of world-wide palaeomagnetic data, Mantles of the Earth and Terrestial Planets*, ed. S. K. Runcorn, *Interscience*, 351.

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(21) Irving, E. & Parry, L. G., 1963. The magnetism of some Permian rocks from New South Wales, *Geophys. J. R. astr. Soc.*, **7**, 395–411.

(22) Irving, E., 1966. Palaeomagnetism of some Carboniferous rocks from New South Wales and its relation to geological events, *J. geophys. Res.*, **71**, 6025–6051.

(23) McElhinny, M. W., 1969. The palaeomagnetism of the Permian of southeast Australia and its significance regarding the problem of intercontinental correlation, *Spec. Publs. geol. Soc. Aust.*, **2**, 61–67.

(24) Azcuy, C. L. & Morelli, J. R., 1970. The Paganzo Basin, Tectonic and sedimentary characteristics of the Gondwana sequences in north-western Argentina. Stratigraphy and Palaeontology of the Gondwana System. Presented to the Second Symposium on Stratigraphy and Palaeontology of the Gondwana System, South Africa (in press).

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